Objectness-Aware Scoring for Few-Shot Video Object Detection

Team Name: Deep Learners — **Members:** Daksh Agarwal (230101123), Kartik Maheshwari (230101116), Aditya Shukla (230101115), Suryadev Vinod (230121059), Sanket Prakash (230103121)

GitHub: `https://github.com/hskad/adaptive-fsvod`

I. Abstract

1. Few-shot video detection must locate new object classes from few labelled examples despite occlusion, motion blur, and changing appearance.
2. Independent frame processing loses useful history; carrying all features forward can propagate background noise.
3. Kumar and Mishra [1] combine pretrained language-aligned visual features, support class prototypes, and attention to confident object features from the previous frame.
4. This selective object memory preserves useful evidence across time while limiting noise, without explicit object tubes.

II. Proposed Novelty & Technical Contribution

1. **Type: algorithmic (inference-time ranking) enhancement**; no architecture, module, or loss change. We apply it to a **reduced adaptation** of [1] (frozen OWLv2-Base features [2], trained temporal head). Objectness-weighted scoring is established [3]; our contribution is its controlled application and evaluation in this few-shot video pipeline.
2. The baseline ranks foreground candidates by class confidence. We multiply this by the frozen encoder's objectness for the same candidate, *before* class-wise non-maximum suppression (NMS). Each candidate's class label, background decision, and refined box remain fixed, as do temporal memory and all learned weights.
3. Validation at seed 42 selects $\beta = 1$ from $\{0, 0.5, 1\}$; this value is then fixed for seeds 43/44 and the fresh test. Both methods share the same checkpoints. No extra training, encoder pass, or trainable parameter is added.

III. Theoretical / Mathematical Justification

1. Let p_j be the maximum foreground softmax score and $q_j \in [0, 1]$ the candidate's class-independent encoder objectness, measured before box refinement. We rank by

$$s_j = p_j q_j^{\beta}, \qquad \log s_j = \log p_j + \beta \log q_j \quad (p_j, q_j > 0).$$

2. For a correct candidate a and an incorrect candidate b with positive scores,

$$s_a > s_b \iff \beta \log(q_a/q_b) > \log(p_b/p_a).$$

   So objectness reverses a wrong class-score ranking when it separates correct from incorrect candidates more strongly than p. Over all pre-NMS foreground candidates of the three models (correct: IoU ≥ 0.5 with a same-class box), it does: the mean log gap (correct − incorrect) is 1.94 for q vs. 0.10 for p, and AUROC is 0.85 vs. 0.68 on validation (test: 0.89 vs. 0.77). AP depends only on score order, so this better ordering raises AP directly.
3. $\beta = 0$ recovers the baseline exactly. Weak objectness can also penalize true objects, so gains are not guaranteed; the product is a ranking heuristic, not a calibrated joint probability.

IV. Comparative Results & Empirical Evaluation

1. **Setup:** custom FSYTV-40 subset from YouTube-VIS 2019; train/validation: 6/3 classes, 18/9 videos, 131/50 frames. Frozen OWLv2-Base uses native 960×960 padded inputs and 128 candidates. Each head trains for 300 updates (AdamW, 10^{-4} learning rate); validation selects steps 250/250/300 for seeds 42/43/44.
2. **Fresh test:** six classes unused in earlier studies, 24 videos, 192 distinct frames; 120 balanced 2-way 1-shot episodes (960 episode-frame instances). Support and query videos differ. Models, β, episodes, and shared features were fixed before scoring.
3. Both methods use the same candidate boxes, score floor 0.001, NMS IoU 0.5, and at most 100 detections per frame. Report frame-level COCO box AP at IoU 0.50:0.05:0.95, AP50, and AP75.

Table 1: Comparison on the fresh six-class subset. AP: mean over three training seeds ± sample SD. Latency: CPU time per frame for heads and decoding (shared encoder excluded). Best AP scores are bold.

Method	AP (mean ± SD, ↑)	AP50 (↑)	AP75 (↑)	Params (M)	Latency (ms ↓)
Reduced baseline ($\beta = 0$)	47.44 ± 0.09	62.44	50.85	2.104	9.22
Objectness score ($\beta = 1$)	$\mathbf{58.30 \pm 0.18}$	**75.04**	**62.80**	2.104	9.19
Δ (Ours – Baseline)	+10.86	+12.61	+11.95	0	−0.03

Result Interpretation: The paired mean gain is **10.86 AP points**; all three seed pairs improve. A 1,000-draw paired, class-stratified query-video bootstrap gives a 95% interval of **[+5.96, +14.85]**, conditional on these classes, fixed support episodes, and trained seed pairs. Bear contributes most of the gain (+43.55 class AP from a 15.10 baseline); **excluding bear, the gain is +4.32 AP**, and all six classes improve. Since weights are shared, the change comes from re-ranking and its downstream filtering, not new feature learning. Latency is unchanged within timing noise.

References:

[1] Kumar and Mishra, *Temporal Object-Aware Vision Transformer for Few-Shot Video Object Detection*, AAAI 2026.
[2] Minderer et al., *Scaling Open-Vocabulary Object Detection*, NeurIPS 2023.
[3] Redmon et al., *You Only Look Once: Unified, Real-Time Object Detection*, CVPR 2016.